

TrES-5b

R-2429

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_140607 · created 2026-08-22T00:25:21+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456815.8164 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.184 +/- 0.051
Transit depth (Rp/Rs)^2	3.39 +/- 1.87 [%]
Orbital Inclination (inc)	81.01 +/- 2.92
Airmass coefficient 1 (a1)	1.161 +/- 0.07
Airmass coefficient 2 (a2)	-0.102 +/- 0.081
Scatter in the residuals of the lightcurve fit is	6.13 %
Best Comparison Star	None
Optimal Aperture	2.29
Optimal Annulus	10.58
Transit Duration (day)	0.046 +/- 0.031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.184 ± 0.051 vs 0.142 (deviation 0.78σ)
Duration fit vs published	0.046 d vs 0.0758 d (deviation 0.91σ)
Mid-time O–C	3.9 min
Depth SNR	3.4
Residual scatter	6.13 %

Light curve

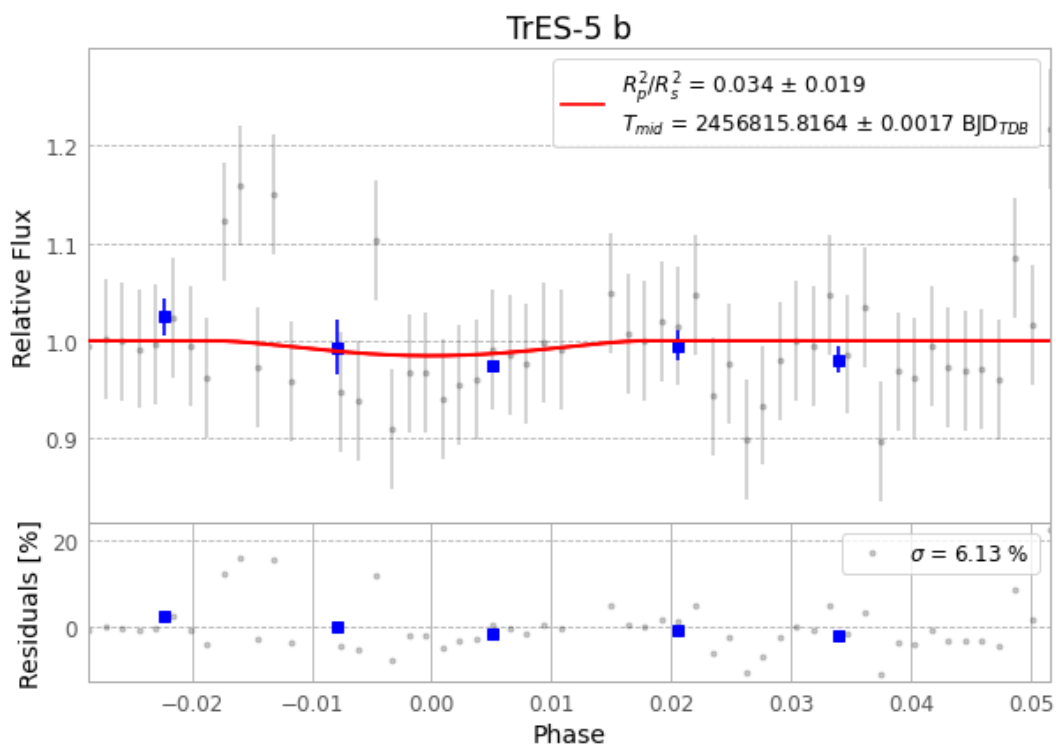


Figure 1 — FinalLightCurve_TrES-5 b_2014-06-06.png (SHA-256 7bdff69465291574...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [365, 249], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 fc3ecb134910831e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

56 FITS frames (37 MB), TRES-5140607063312.FITS → TRES-5140607092413.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-140607.log (d1bf065b5f98...), FinalLightCurve_TrES-5 b_2014-06-06.png (7bdff6946529...), FinalLightCurve_TrES-5 b_2014-06-06.csv (3e52d3833f84...)

Compute resources

Wall time 107.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-22 00:25Z.